

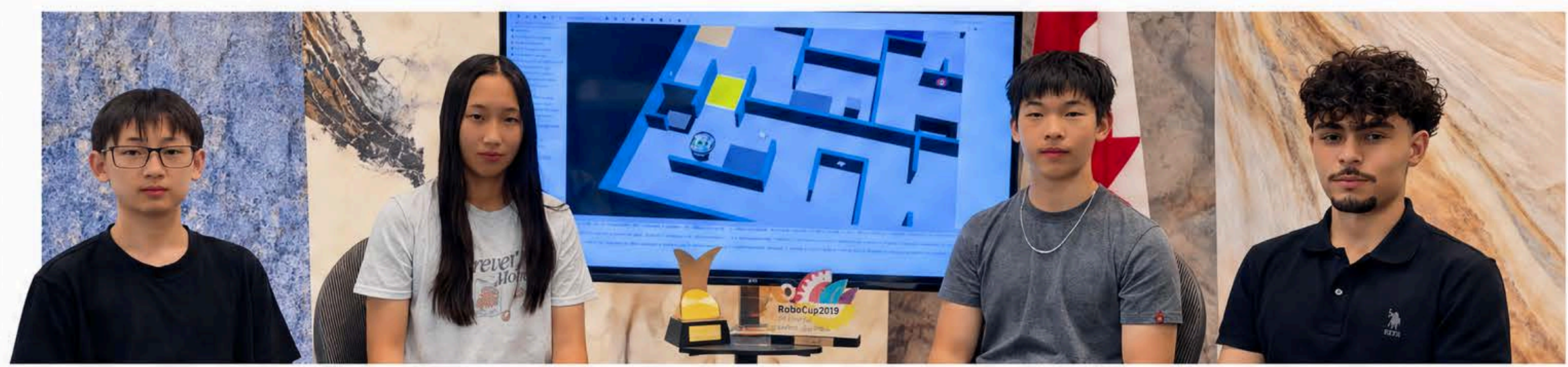
Kavosh_Simul

RoboCupJunior Rescue Simulation 2026 | Canada |
Kavosh AI & Robotics Academy

Lightweight SLAM • Wall-Based Search • Team-Trained Vision



Experience in FIRA, VEX, RoboCup Vancouver 2026 qualifiers.



Sam Liu

Mapping + SLAM

Lightweight Python SLAM,
wall-side mapping,
LiDAR pattern logic.

Zhengyu "Victoria" Wu

Navigation Logic

Movement decisions,
recovery, wall-focused
exploration.

Hanwen "Henry" Yang

Movement + Path Planning

Turning, wall alignment,
obstacle response,
A*-style planning.

Ermia Karimi

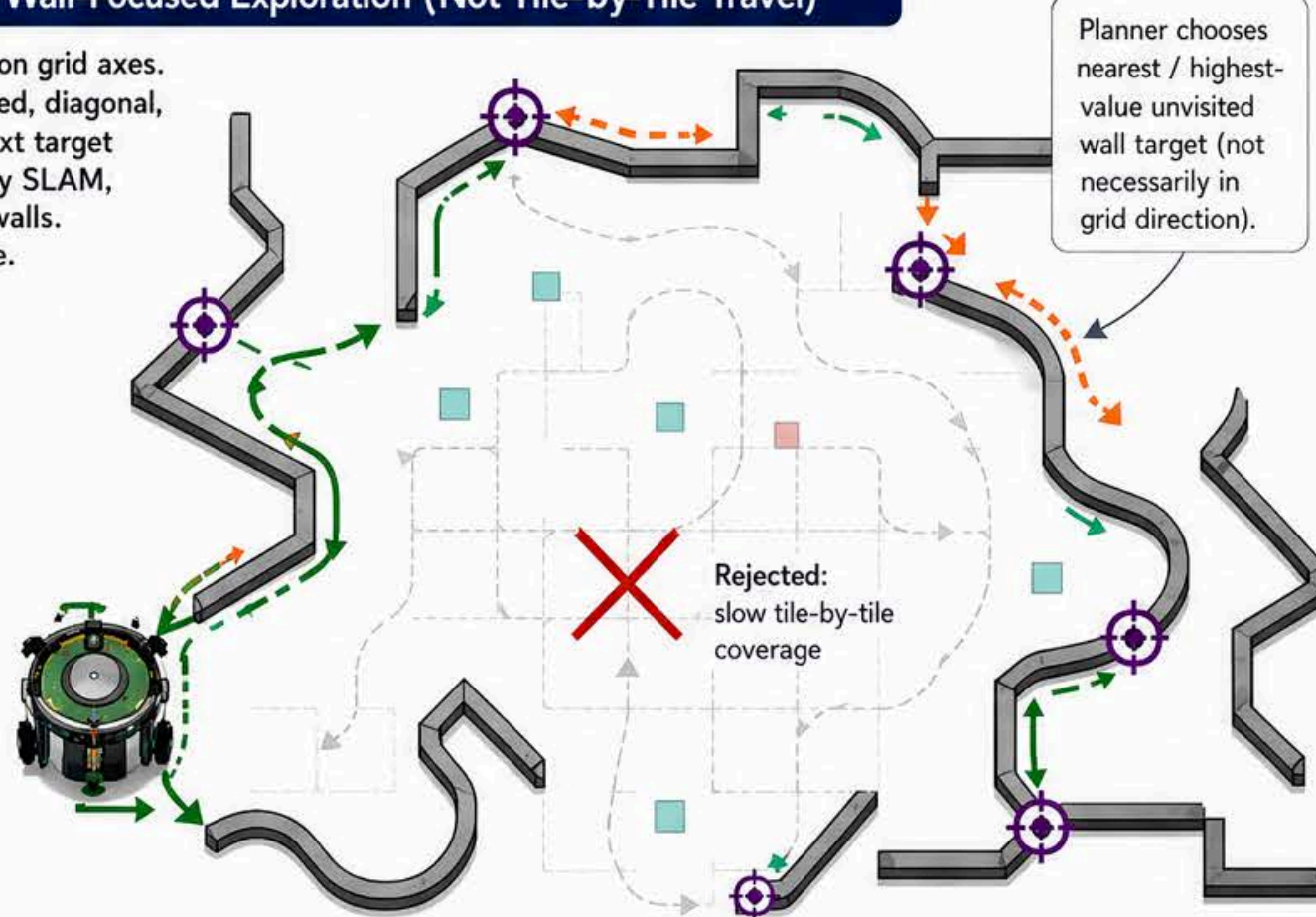
Computer Vision

ROI pipeline, team-trained
YOLO11s, OpenCV
cognitive targets.

1. Wall-Focused Exploration (Not Tile-by-Tile Travel)

The planner is not limited to moving on grid axes. It freely follows any wall shape (curved, diagonal, angled, irregular) and selects the next target based on unvisited walls detected by SLAM, because tokens and victims are on walls. This reduces unnecessary travel time.

- Planned wall-adjacent path (free movement)
- Unvisited wall targets (selected by planner)
- High-value wall target (token/victim likely)
- Rejected: tile-by-tile grid coverage
- Black hole / unsafe cell
- Floor colour tag (on unvisited tile)



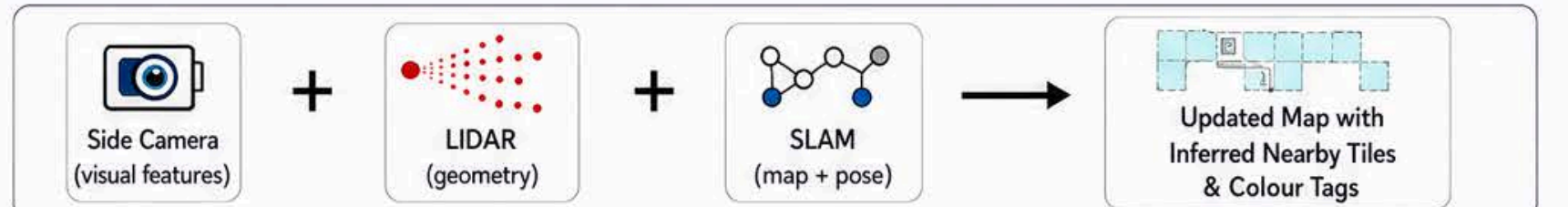
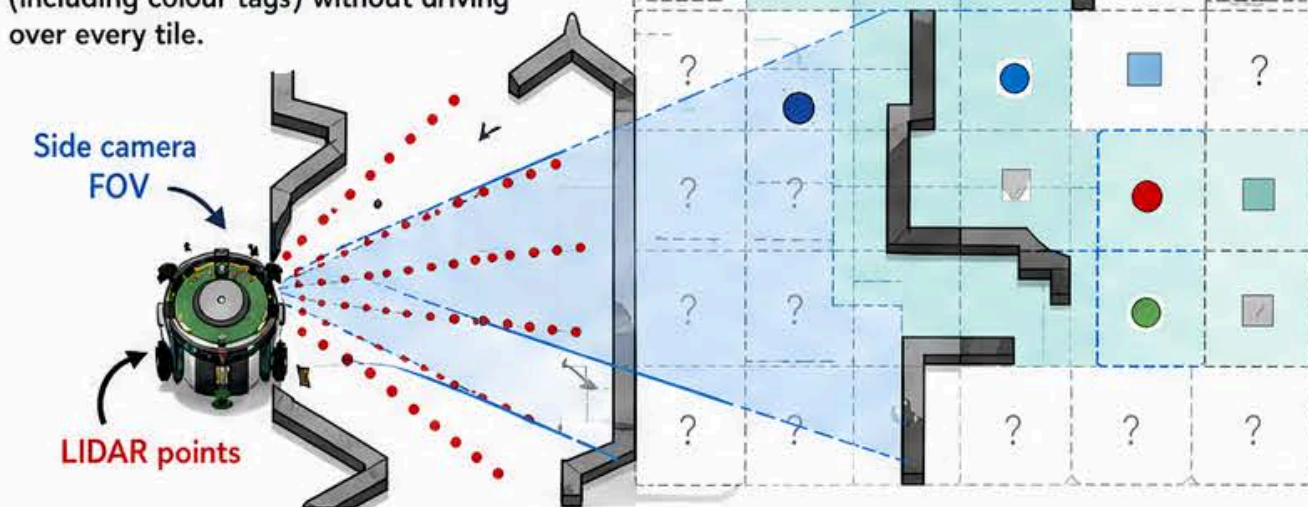
Planner chooses nearest / highest-value unvisited wall target (not necessarily in grid direction).

Rejected: slow tile-by-tile coverage

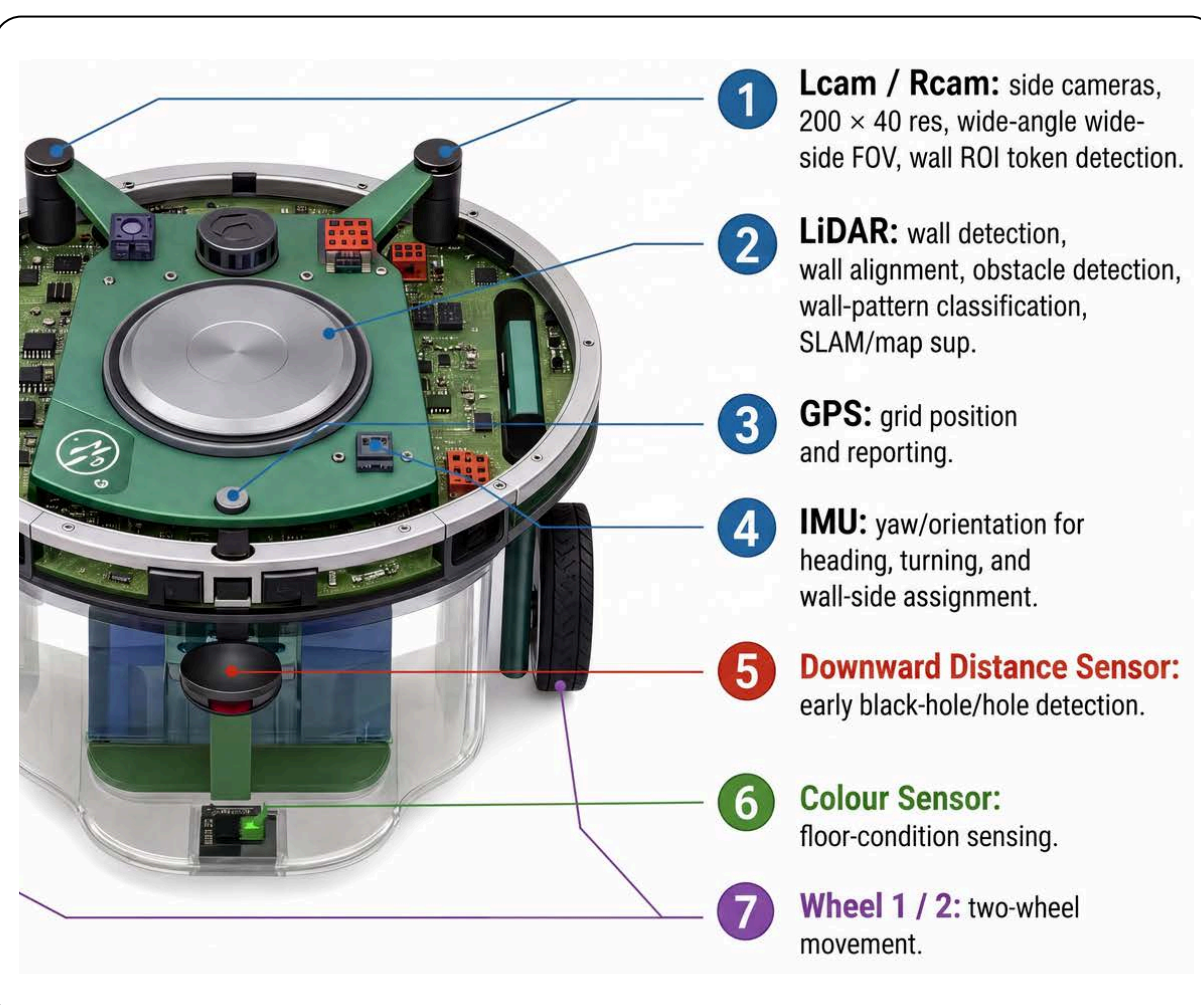
Result: Faster discovery of wall tokens with fewer redundant moves.

2. Vision + SLAM Mapping of Unvisited Tiles and Wall Features

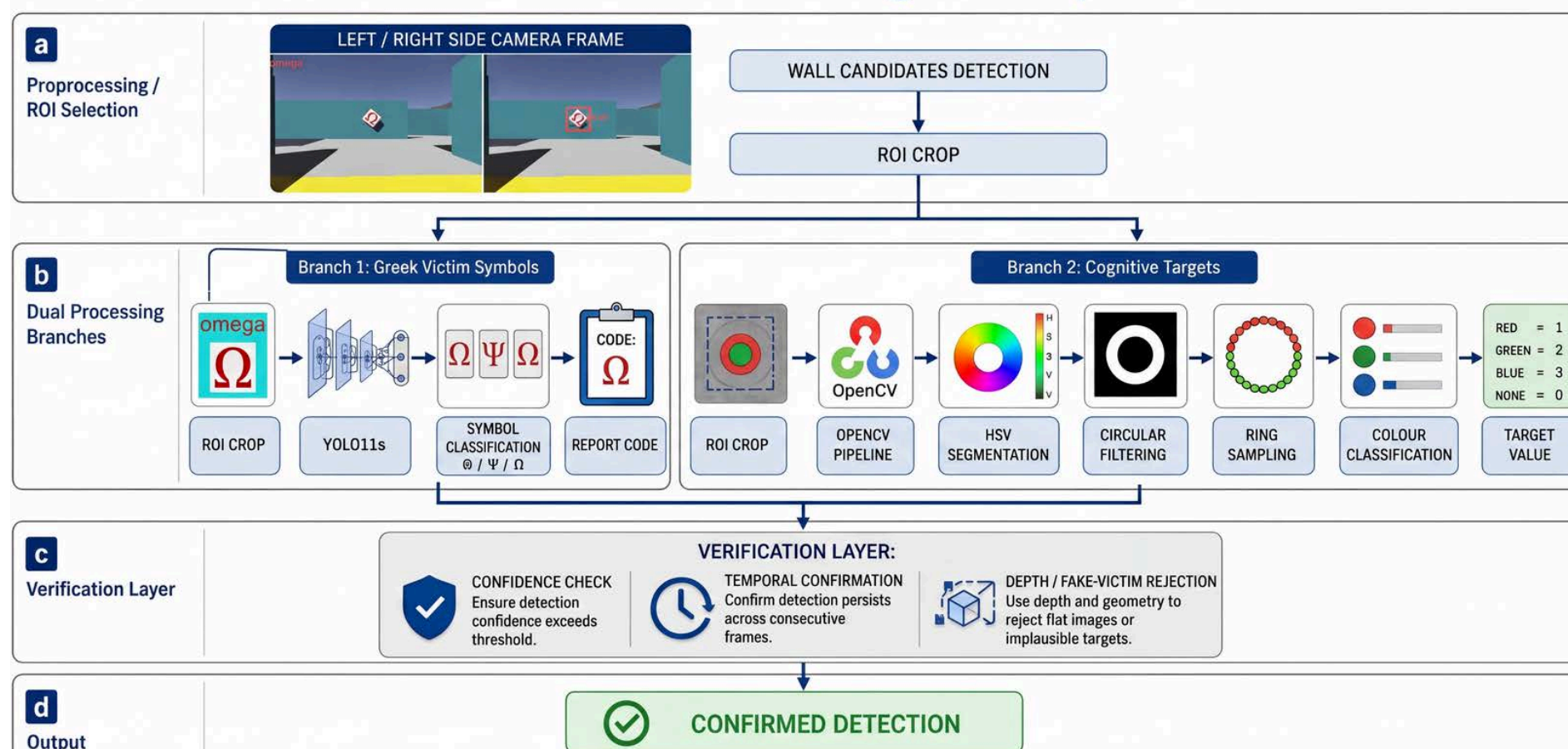
Side cameras and LiDAR observe walls and nearby areas. SLAM + image processing infer and fill unvisited tiles and wall features (including colour tags) without driving over every tile.



Result: Map nearby unvisited tiles and colour tags without entering them.



ROI-Based Vision Integration Pipeline



Victim and Wall-Token Detection Results

Precision	98.39%
Recall	91.99%
F1	95.08%
mAP50	97.45%
mAP50-95	92.36%

Stress audit: ~3,000 images

- ✓ 36/36 victim detections
- ✓ 0 hard-negative false positives